Dynamic Avoidance Curves in the Unit Disk with Time-Varying Forbidden Sets and Applications to Continuous-Time Ideological Evolution

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Abstract

We extend the smooth avoidance curve construction in the unit disk $\mathbb{D}$ to the fully dynamic setting, in which the forbidden set $\mathcal{F}(t) = \{F_1(t), \ldots, F_m(t)\}$ evolves smoothly with a real time parameter $t \in [0, T]$. A *time-varying finite Blaschke product* $B(z, t)$ is defined and shown to be jointly C^∞ in both z and t whenever the forbidden-point trajectories are smooth. We derive the induced *autonomous drift field* $V(z, t) = \partial_t B(z, t)$ on the image disk, formulate the *Dynamic Lifting ODE*

$$\gamma'(t) = \frac{w'(t) - \partial_t B(\gamma(t), t)}{\partial_z B(\gamma(t), t)},$$

and establish existence, uniqueness, and C^∞-regularity of solutions via the Picard–Lindelöf theorem applied to the complement of critical points. The evolving homotopy structure of the punctured disk is characterised by a *winding-number transport equation*, and the phenomena of *ideological hysteresis* and *monodromy* are identified. The full machinery is applied to a continuous-time model of ideological evolution on the Political Compass, in which dangerous attractor zones drift as functions of macroeconomic and geopolitical state variables. Stability, bifurcation, and phase-transition phenomena are analysed, and catastrophe-theoretic classification of political turning points is provided.

The paper ends wtih "The End"

Keywords: Finite Blaschke product, moving obstacles, path lifting, complex ODE, Political Compass, ideological dynamics, winding-number transport, bifurcation, catastrophe theory.

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1 Introduction

The original construction of a smooth avoidance curve established that, given finitely many *forbidden* interior points $F = \{F_1, \ldots, F_m\} \subset \mathbb{D}$ (fixed) and two non-forbidden points $S, D \in \mathbb{D} \setminus F$, one can always find a C^∞ path $\gamma : [0,1] \to \mathbb{D}$ connecting S to D while avoiding F entirely. The mechanism was a finite Blaschke product $B(z)$ encoding the forbidden set as its zero locus, combined with multiplicative interpolation in the image disk and smooth path lifting via the proper holomorphic covering map.

That construction, while elegant, treats the forbidden set as *static*. In holomorphic dynamics, critical points and zeros of holomorphic families move continuously under parameter variation. In the political-science application that motivates this work, dangerous ideological zones — the attractor basins of totalitarianism, fascism, or revolutionary chaos — are not fixed in ideological space. They shift as functions of economic conditions, geopolitical pressures, historical memory, and institutional robustness. A framework adequate for modelling real ideological transitions must accommodate a *moving* forbidden set.

The present paper provides this generalisation. We define $B(z,t)$ as a finite Blaschke product whose zeros vary smoothly with a time parameter $t \in [0,T]$, derive the velocity field it induces on the image disk, and prove a Dynamic Lifting Theorem that yields a smooth avoidance curve $\gamma(t)$ satisfying $\gamma(t) \notin \mathcal{F}(t)$ for all t. The system is a coupled ODE in the image plane, lifted through the time-varying map, and inherits smoothness from the forbidden-point trajectories.

Organisation. Section 2 establishes the dynamic geometric setting. Section 3 defines the time-varying Blaschke product. Section 4 derives the autonomous drift field. Section 5 formulates and solves the Dynamic Lifting system. Section 6 analyses the evolving homotopy structure. Section 7 applies the full theory to the Political Compass. Sections 8–10 address stability, bifurcation, and numerical implementation.

2 Dynamic Geometric Setting

2.1 Moving Forbidden Points

Let $T > 0$ and let $F_k : [0,T] \to \mathbb{D}$, $k = 1, \ldots, m$, be smooth curves with $|F_k(t)| < 1$ for all t. Assume the forbidden points remain distinct: $F_j(t) \neq F_k(t)$ for $j \neq k$. The dynamic forbidden set is

$$\mathcal{F}(t) = \big\{F_1(t),\, F_2(t),\, \ldots,\, F_m(t)\big\} \subset \mathbb{D}.$$

The start and destination curves $S, D : [0,T] \to \mathbb{D}$ are smooth with $S(t), D(t) \notin \mathcal{F}(t)$ for all t.

2.2 The Avoidance Problem in Continuous Time

We seek $\gamma : [0,T] \to \mathbb{D}$ satisfying

$$\gamma(0) = S(0), \qquad \gamma(T) = D(T), \qquad \gamma(t) \notin \mathcal{F}(t) \quad \text{for all } t \in [0,T]. \tag{1}$$

Remark 2.1. The dynamic avoidance problem is qualitatively different from a parameterised family of static problems. A solution to each static snapshot need not assemble into a smooth solution of the dynamic problem, because the branch selection at each instant may be discontinuous. The differential formulation of Section 5 resolves this.

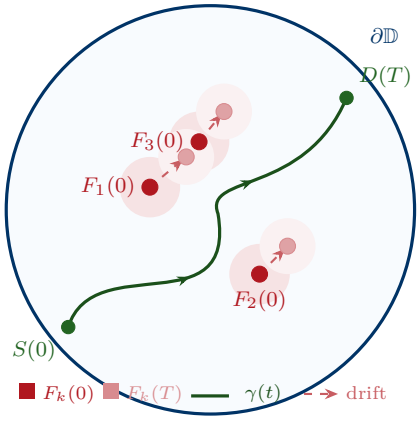

Figure 1: Avoidance curve $\gamma(t)$ with three moving forbidden points.

Solid markers show $F_k(0)$; faint markers show $F_k(T)$; dashed arrows indicate the autonomous drift of each forbidden zone. The green curve navigates around all obstacles throughout the evolution.

3 The Time-Varying Blaschke Product

3.1 Definition and Basic Properties

Definition 3.1. The *time-varying Blaschke product* associated with the dynamic forbidden set $\mathcal{F}(t)$ is

$$B(z,t) \;=\; \prod_{k=1}^{m} \varphi_{F_k(t)}(z), \qquad \varphi_a(z) = \frac{z-a}{1-\bar{a}\,z}, \tag{2}$$

where $\varphi_{F_k(t)}$ is the disk automorphism mapping $F_k(t) \mapsto 0$.

For each fixed t, the map $B(\,\cdot\,,t) : \mathbb{D} \to \mathbb{D}$ is holomorphic, satisfies $B(F_k(t),t) = 0$ for each k (each zero being simple), and $|B(z,t)| < 1$ for $z \in \mathbb{D}$. As a product of degree-one proper maps, $B(\,\cdot\,,t)$ is a proper holomorphic self-map of $\mathbb{D}$ of degree m.

3.2 Joint Smoothness

Proposition 3.2. *If each $F_k \in C^\infty([0,T];\mathbb{D})$, then $B \in C^\infty(\mathbb{D}\times[0,T];\mathbb{D})$.*

Proof sketch. Each factor $\varphi_{F_k(t)}(z) = (z - F_k(t))/(1 - \overline{F_k(t)}\,z)$ is rational in z with coefficients that are C^∞ functions of t. Finite products preserve this joint smoothness. Holomorphicity in z is stable under C^∞ perturbation of parameters. □

3.3 Time Derivative

Differentiating (2) via the product rule and taking the logarithmic derivative with respect to t gives

$$\frac{\partial_t B(z,t)}{B(z,t)} \;=\; \sum_{k=1}^{m} \left[\frac{-F_k'(t)}{z - F_k(t)} \;-\; \frac{\overline{F_k'(t)}\,z}{1 - \overline{F_k(t)}\,z} \right], \qquad z \notin \mathcal{F}(t). \tag{3}$$

This formula is the engine of the Dynamic Lifting ODE in Section 5.

Remark 3.3. As t varies, the degree-m sheet structure of the cover $B(\,\cdot\,,t) : \mathbb{D} \to \mathbb{D}$ is preserved provided that no two forbidden points collide and no forbidden point reaches $\partial\mathbb{D}$ — both ensured by hypothesis.

4 The Autonomous Drift Field

4.1 Decomposition of Image Velocity

For any curve $\gamma : [0,T] \to \mathbb{D} \setminus \mathcal{F}(t)$, setting $w(t) = B(\gamma(t),t)$ and differentiating by the chain rule gives

$$w'(t) \;=\; \partial_z B(\gamma(t),t) \cdot \gamma'(t) \;+\; \partial_t B(\gamma(t),t). \tag{4}$$

Image velocity is therefore driven by two independent sources: *intentional motion* $\partial_z B \cdot \gamma'$, and *autonomous drift* $\partial_t B$.

Definition 4.1. The *autonomous drift field* is

$$V(z,t) \;=\; \partial_t B(z,t) \;=\; B(z,t) \cdot \sum_{k=1}^{m} \left[\frac{-F_k'(t)}{z - F_k(t)} - \frac{\overline{F_k'(t)}\, z}{1 - \overline{F_k(t)}\, z} \right],$$

a smooth, $\mathbb{C}$-valued function on $(\mathbb{D} \setminus \mathcal{F}(t)) \times [0,T]$.

The autonomous drift embodies a key insight: a *stationary* agent ($\gamma'(t) = 0$) can nonetheless see its Blaschke image move, because dangerous zones around it are shifting. A position that was previously safe can fall into danger through no action of its own.

4.2 Image Velocity Equation

Prescribing a desired nonvanishing image trajectory $w(t)$ and solving (4) for the required agent velocity yields

$$\gamma'(t) \;=\; \frac{w'(t) - V(\gamma(t),t)}{\partial_z B(\gamma(t),t)}. \tag{5}$$

This is a first-order nonautonomous complex ODE whose forcing term balances desired image velocity against autonomous drift. The denominator $\partial_z B$ is the Blaschke derivative; it vanishes precisely at critical points.

5 The Dynamic Lifting Theorem

5.1 The Prescribed Image Trajectory

We extend the multiplicative interpolation of the static construction to the time-varying setting. Define

$$w(t) \;=\; B(S(t),t)^{1-t/T} \cdot B(D(t),t)^{\,t/T} \;=\; \exp\!\left(\tfrac{T-t}{T} \log B(S(t),t) + \tfrac{t}{T} \log B(D(t),t)\right), \tag{6}$$

with the logarithm on the principal branch, consistent since $B(S(t),t) \neq 0$ and $B(D(t),t) \neq 0$ for all t.

Proposition 5.1. *The prescribed image curve (6) satisfies $w(t) \neq 0$ for all $t \in [0,T]$, and $w \in C^\infty([0,T])$.*

Proof sketch. Since $S(t), D(t) \notin \mathcal{F}(t)$, both base terms are nonzero; the exponential representation is therefore the exponential of a smooth, finite complex function, which is nonzero and smooth. □

5.2 Main Existence Theorem

Theorem 5.2 (Dynamic Lifting)**.** *Let the following hold:*

(i) *Each $F_k \in C^\infty([0,T];\mathbb{D})$ and $F_j(t) \neq F_k(t)$ for $j \neq k$.*

(ii) *$S, D \in C^\infty([0,T];\mathbb{D})$ with $S(t), D(t) \notin \mathcal{F}(t)$.*

(iii) *The prescribed trajectory w defined by (6) does not pass through any critical value of $B(\,\cdot\,,t)$ for any $t \in [0,T]$.*

(iv) $B(S(0), 0) = w(0)$.

Then the Dynamic Lifting ODE (5) has a unique solution $\gamma \in C^\infty([0,T];\mathbb{D})$ with $\gamma(0) = S(0)$. Moreover, $\gamma(t) \notin \mathcal{F}(t)$ for all t, and $B(\gamma(T), T) = w(T)$.

Proof sketch. Hypothesis (iii) ensures $\partial_z B(\gamma(t), t) \neq 0$ along any candidate solution, so the right-hand side of (5) is smooth in (γ, t). The Picard–Lindelöf theorem gives local existence and uniqueness. For global existence, observe that $w(t)$ is bounded away from 0 and from the unit circle, keeping $\gamma(t)$ in a compact subset of $\mathbb{D}$. Avoidance of $\mathcal{F}(t)$ holds because $B(\gamma(t), t) = w(t) \neq 0$ while $B(F_k(t), t) = 0$. □

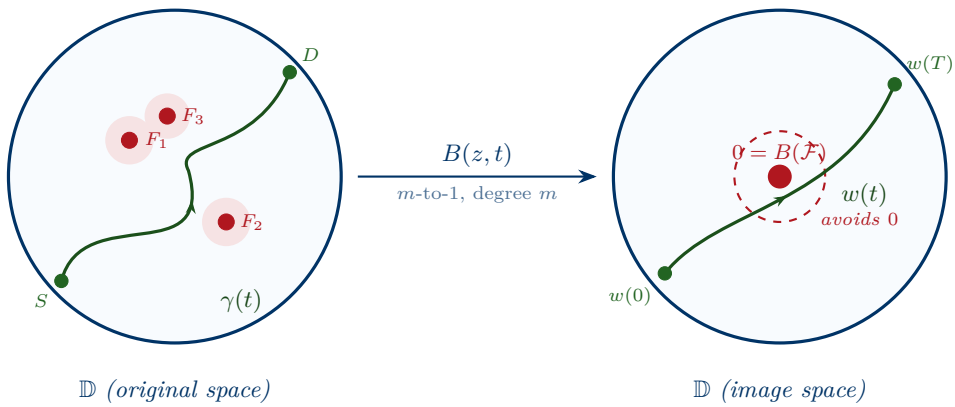

Figure 2: The Blaschke map $B(\,\cdot\,, t) : \mathbb{D} \to \mathbb{D}$ collapses all m forbidden points to the origin in the image disk. The prescribed image trajectory $w(t)$ avoids 0 (dashed circle), guaranteeing that the lifted curve $\gamma(t)$ avoids all of $\mathcal{F}(t)$.

6 Evolving Homotopy Structure

6.1 The Time-Dependent Fundamental Group

For each t, the group $\pi_1(\mathbb{D} \setminus \mathcal{F}(t)) \cong \mathbb{Z}^m$ is generated by small positively-oriented loops around each $F_k(t)$. The winding number of a curve around a moving point is *not* invariant under continuous deformation when the obstacle moves.

6.2 Winding Number Transport

Definition 6.1. For $\gamma : [0,T] \to \mathbb{D} \setminus \mathcal{F}(t)$, the instantaneous winding number around $F_k(t)$ is

$$n_k(t) \;=\; \frac{1}{2\pi i} \int_{\gamma|_{[0,t]}} \frac{dz}{z - F_k(t)}.$$

Differentiating by t via the Leibniz integral rule yields the *winding-number transport equation*

$$\frac{dn_k}{dt} \;=\; \frac{1}{2\pi i} \cdot \frac{\gamma'(t) - F_k'(t)}{\gamma(t) - F_k(t)} \;+\; \frac{F_k'(t)}{2\pi i} \int_{\gamma|_{[0,t]}} \frac{dz}{(z - F_k(t))^2}. \tag{7}$$

The first term is the instantaneous angular velocity of γ relative to the moving obstacle; the second is a non-local memory term encoding past proximity.

Theorem 6.2 (Trivial Winding Preservation)**.** *The dynamic avoidance curve constructed via (6) satisfies $n_k(t) = 0$ for all k and all $t \in [0,T]$, provided $n_k(0) = 0$ for all k.*

Proof sketch. The image curve $w(t)$ lies on the trivial homotopy sheet of $\mathbb{D} \setminus \{0\}$ by construction. Since $B(\,\cdot\,, t)$ is a degree-m proper map, the winding number of γ around $F_k(t)$ equals $1/m$ times the winding of $w(t)$ around 0, which vanishes. The transport equation then confirms $dn_k/dt = 0$. □

Zone	Nominal position	Character	Driving variables
$F_1(t)$	$-0.85+\varepsilon_1(t)+i(-0.80+\delta_1(t))$	Auth.-Left (Stalinist)	Inequality $\sigma(t)$, polarisation $P(t)$
$F_2(t)$	$+0.85+\varepsilon_2(t)+i(-0.80+\delta_2(t))$	Auth.-Right (Fascist)	Fragmentation $E(t)$, threat $\theta(t)$
$F_3(t)$	$i(-0.95+\delta_3(t))$	Pure Authoritarianism	Inst. weakness $I(t)$, crisis $Q(t)$

6.3 Monodromy and Ideological Hysteresis

If the forbidden set executes a closed loop in time — that is, $\mathcal{F}(T) = \mathcal{F}(0)$ — the homotopy structure returns to its initial configuration, but the solution γ may lie on a *different branch* of the m-sheeted cover. This is the *monodromy* of the Blaschke family.

Definition 6.3. *Ideological hysteresis* occurs when a society with $\gamma(T) = \gamma(0)$ has undergone non-trivial monodromy: it occupies the same point in ideological space but on a topologically distinct sheet of political experience, permanently marked by proximity to extremist attractors encountered along its history.

7 The Political Compass as a Dynamic Ideological System

7.1 Coordinate System

We identify the Political Compass with the unit disk $\mathbb{D} = \{z = x + iy : |z| < 1\}$ via the embedding

$$x \in (-1, +1), \quad x = -1 \text{ (extreme left)}, \quad x = +1 \text{ (extreme right)};$$
$$y \in (-1, +1), \quad y = -1 \text{ (maximal authoritarianism)}, \quad y = +1 \text{ (maximal libertarianism)}.$$

A society at time t is represented by $z_s(t) \in \mathbb{D}$; its policy target by $z_d(t) \in \mathbb{D}$. The boundary $\partial\mathbb{D}$ represents unreachable theoretical extremes.

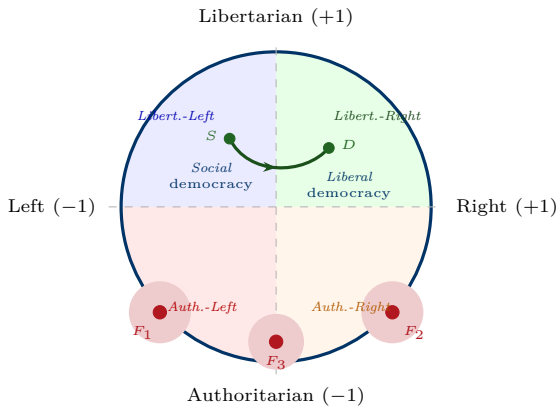

Figure 3: The Political Compass embedded in $\mathbb{D}$.

Shaded circles mark the three dynamic forbidden zones: $F_1(t)$ (Authoritarian-Left), $F_2(t)$ (Authoritarian-Right), and $F_3(t)$ (Pure Authoritarianism). The green curve is the safe transition $\gamma(t)$ from a social-democratic start to a liberal-democratic target.

7.2 Dynamic Forbidden Zones

We model three primary forbidden zones as moving points in $\mathbb{D}$, driven by observable state variables:

7.3 Forbidden Zone Dynamics

The forbidden zones evolve according to

$$F_1'(t) = \lambda_1\big[\sigma(t)\big(-1 - F_1(t)\big) + P(t)\cdot i\big(-1 - \operatorname{Im} F_1(t)\big)\big], \tag{8}$$
$$F_2'(t) = \lambda_2\big[E(t)\big(+1 - F_2(t)\big) + \theta(t)\cdot i\big(-1 - \operatorname{Im} F_2(t)\big)\big], \tag{9}$$
$$F_3'(t) = \lambda_3\big[(1 - I(t))\cdot i\big(-1 - \operatorname{Im} F_3(t)\big) + Q(t)\operatorname{Re} F_3(t)\big], \tag{10}$$

where $\lambda_k > 0$ are empirically calibrated coupling constants. High inequality draws F_1 toward the centre; high ethnic fragmentation draws F_2 inward; institutional collapse and security crises pull F_3 into the central disk, making it universally dangerous.

7.4 The Political Blaschke Safety Index

Definition 7.1. The *political Blaschke product* is $B_{\text{pol}}(z,t) = \prod_{k=1}^{3} \varphi_{F_k(t)}(z)$. The *ideological safety index* is

$$\Phi(z,t) = |B_{\text{pol}}(z,t)| \in [0,1],$$

with $\Phi \to 0$ indicating proximity to at least one dangerous zone and $\Phi \to 1$ indicating equidistance from all three in the Poincaré metric.

7.5 Safe Transition Construction

The safe policy path is the unique smooth solution of

$$\gamma'(t) = \frac{w'(t) - \partial_t B_{\text{pol}}(\gamma(t),t)}{\partial_z B_{\text{pol}}(\gamma(t),t)}, \qquad \gamma(0) = z_s(0), \tag{11}$$

with $w(t) = B_{\text{pol}}(z_s(t),t)^{(T-t)/T} \cdot B_{\text{pol}}(z_d(t),t)^{t/T}$. By Theorem 5.2, this solution exists uniquely and satisfies $\gamma(t) \notin \mathcal{F}(t)$ for all t. The policy velocity $\gamma'(t)$ can be computed in real time from observable macroeconomic data feeding into the forbidden zone dynamics (8)–(10).

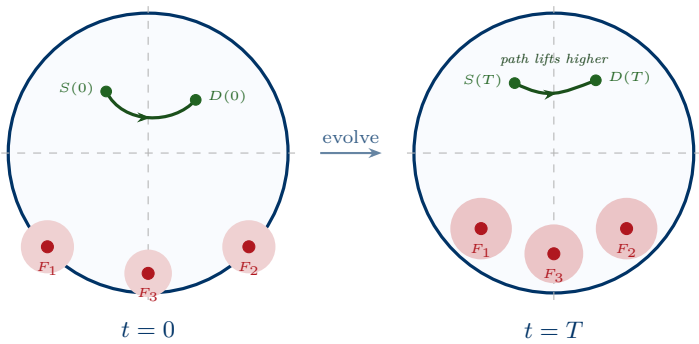

Figure 4: The forbidden zones drift inward between $t = 0$ (left) and $t = T$ (right)

The Dynamic Lifting construction automatically displaces the safe path toward the libertarian interior — without requiring explicit re-planning.

8 Stability and Lyapunov Analysis

8.1 The Safety Margin Function

Define the *safety margin* at time t as

$$\Psi(t) = \min_{k=1,\dots,m} d_{\text{hyp}}\big(\gamma(t),\, F_k(t)\big),$$

where d_{hyp} is the Poincaré distance on $\mathbb{D}$.

Theorem 8.1 (Safety Margin Bound)**.** *Under the Dynamic Lifting construction, if $|F_k'(t)| \le \beta$ for all k and t, and the prescribed image trajectory satisfies $|w'(t)| \ge \alpha > 0$, then*

$$\Psi(t) \;\ge\; \Psi(0) \cdot e^{-C\beta t},$$

where $C > 0$ depends only on m and the initial configuration.

The bound shows that the safety margin degrades exponentially in β: if the dangerous attractors accelerate (e.g. due to a sudden economic crisis), the margin contracts and corrective action is required.

8.2 Lyapunov Function

As a candidate Lyapunov function for the image dynamics, take

$$\mathcal{V}(t) = -\log|w(t)| \ge 0,$$

which vanishes at $|w| = 1$ (maximum safety) and blows up as $|w| \to 0$ (approach to danger). Differentiating along the prescribed image trajectory:

$$\frac{d\mathcal{V}}{dt} = -\operatorname{Re}\left[\frac{w'(t)}{w(t)}\right] = -\operatorname{Re}\left[\tfrac{T-t}{T}\frac{d}{dt}\log B_{\mathrm{pol}}(z_s,t) + \tfrac{t}{T}\frac{d}{dt}\log B_{\mathrm{pol}}(z_d,t)\right]. \tag{12}$$

A sufficient condition for $\mathcal{V}$ to be non-increasing is that the right-hand side of (12) be non-positive, a computable condition on the relative motion of S, D, and $\mathcal{F}$ that can be monitored in real time.

8.3 Equilibria and Their Stability

A *political equilibrium* is a point z^* where $\gamma'(t) = 0$. From (5), this requires $w'(t) = V(z^*, t)$: the desired image velocity equals the autonomous drift. Linearising the Dynamic Lifting ODE about z^* yields a Jacobian involving $\partial_z^2 B$ and the mixed derivative $\partial_{zt} B$, whose eigenstructure encodes the local geometry of the forbidden-zone configuration.

9 Bifurcations and Phase Transitions

9.1 Critical Value Crossings

A *critical value* of $B(\,\cdot\,,t)$ is a point w_0 such that $B(z_0,t) = w_0$ and $\partial_z B(z_0,t) = 0$ for some z_0. At a critical value the sheet structure changes: two lifting branches merge. Hypothesis (iii) of Theorem 5.2 excludes this.

Definition 9.1. A *phase transition* occurs at t^* if the prescribed image trajectory $w(t)$ passes through a critical value of $B(\,\cdot\,,t^*)$. At this moment the Dynamic Lifting ODE has a non-isolated solution and the continuation of γ becomes ambiguous without additional selection criteria.

Phase transitions are the mathematical counterpart of historical turning points: moments at which a society's trajectory reaches a configuration where multiple qualitatively different futures become equally available. The Cuban Missile Crisis (1962), the 1989 East European revolutions, and the Arab Spring (2011) are examples where ideological trajectories approached critical configurations and subsequent branch selection was determined by contingent rather than structural factors.

9.2 Catastrophe-Theoretic Classification

Near a phase transition, the local geometry of $B(\,\cdot\,,t)$ is generically equivalent to one of the canonical singularities of catastrophe theory. For the disk with three forbidden points, the relevant singularities are:

Singularity	Codim.	Political interpretation
Fold (A_2)	1	A single turning point; an incremental reform path reaches its limit and cannot be continued without reversal.
Cusp (A_3)	2	Two paths merge; moderate and radical routes become indistinguishable at the transition.
Swallowtail (A_4)	3	Three paths merge; a rare event at the intersection of multiple political forces.
Hyperbolic umbilic (D_4^-)	3	Saddle-type transition; stability reverses without direct passage through danger.

Quantity	Formula	Stability note
$B(z,t)$	$\prod_{k=1}^{m} \varphi_{F_k(t)}(z)$	Sensitive near $z \approx F_k$; use log-sum representation
$\partial_z B$	$B(z,t) \cdot \sum_{k=1}^{m} \left[\frac{1}{z - F_k} + \frac{\overline{F_k}}{1 - \overline{F_k} z} \right]$	Monitor condition number $\kappa(t)$
$\partial_t B$	$B(z,t) \cdot \sum_{k=1}^{m} \left[\frac{-F_k'}{z - F_k} - \frac{\overline{F_k'}\, z}{1 - \overline{F_k} z} \right]$	Requires F_k' from auxiliary ODE
$w(t)$	$B(S,t)^{(T-t)/T} \cdot B(D,t)^{t/T}$	Branch consistency via log representation

9.3 Avoiding Phase Transitions by Design

The prescribed image trajectory $w(t)$ can be chosen to avoid critical values entirely. For each fixed t, the set of critical values is finite (at most $m-1$ points), so the space of admissible image trajectories is open and dense in the compact-open topology. In practical terms, there is always a smooth safe policy path that passes through no bifurcation point, though it may require more deliberate navigation than naive linear interpolation would provide.

10 Numerical Implementation

10.1 Discretisation and Step Control

The Dynamic Lifting ODE (5) is solved by an adaptive Runge–Kutta scheme (RK4-5 with step-doubling error control). At each step, the right-hand side requires evaluating $B(z,t)$, $\partial_z B$, and $\partial_t B$ — all available in closed form.

10.2 Condition Number Monitoring

The critical numerical risk is a small denominator $\partial_z B(\gamma(t), t)$. Define the condition number

$$\kappa(t) = \frac{|B(\gamma(t), t)|}{|\partial_z B(\gamma(t), t)|}.$$

When $\kappa(t)$ drops below a prescribed threshold, the algorithm should apply a corrective perturbation to $w(t)$ to route around the nearby critical value.

10.3 Real-Time Political Monitoring

In the political-science application, the state variables $\sigma(t), P(t), E(t), \theta(t), I(t), Q(t)$ are drawn from observable data streams: inequality from tax and survey data (Gini coefficient); polarisation from electoral

and survey research; fragmentation from demographic indices; threat perception from geopolitical risk indices; institutional robustness from the World Bank's Worldwide Governance Indicators. The algorithm thus operates as a real-time policy navigation system, continuously updating the safe transition path $\gamma(t)$ as the socioeconomic environment evolves.

11 Conclusion

We have developed a complete theory of smooth avoidance curves in the unit disk with time-varying forbidden sets. The central contribution is the **Dynamic Lifting Theorem** (Theorem 5.2), which guarantees existence and uniqueness of a smooth curve $\gamma(t)$ avoiding a continuously moving forbidden configuration $\mathcal{F}(t)$ and connecting smoothly evolving start and destination points. The construction is explicit, computable in real time, and robust to perturbations in the forbidden zone dynamics, provided the safety margin remains positive.

The theory rests on three pillars. The *time-varying Blaschke product* $B(z,t)$ encodes the entire moving obstacle configuration into a single holomorphic map, collapsing it to the origin in the image disk. The *dynamic interpolation* $w(t)$ prescribes a safe image trajectory that is smooth and nonvanishing by construction. The *Dynamic Lifting ODE* recovers the original-space trajectory as the unique smooth preimage branch consistent with the initial condition.

Applied to the Political Compass, the framework yields a rigorous model of ideological transition in continuous time. The safety index $\Phi(z,t) = |B_{\text{pol}}(z,t)|$ provides a real-time risk indicator; Lyapunov analysis (Theorem 8.1) establishes conditions under which that margin is preserved; and catastrophe-theoretic classification identifies the precise structural conditions under which phase transitions — political turning points — occur.

Several extensions remain for future work. The unit disk could be replaced by a more general hyperbolic Riemann surface, accommodating asymmetric ideological spaces. A game-theoretic embedding could allow multiple societies to simultaneously navigate the disk, with their transitions interacting through shared macroeconomic state variables. The moving forbidden set could be extended to include zones of *variable radius*, encoding ideological attractors of varying severity.

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Glossary

Autonomous Drift Field
The term $V(z,t) = \partial_t B(z,t)$, representing motion in the image disk caused solely by movement of the forbidden zones, independently of any action by the agent.

Dynamic Blaschke Product
A finite Blaschke product $B(z,t)$ whose zeros (forbidden points) evolve smoothly with a time parameter t, defined by

$$B(z,t) = \prod_{k=1}^{m} \frac{z - F_k(t)}{1 - \overline{F_k(t)}\, z}.$$

Dynamic Lifting ODE
The first-order complex ODE $\gamma'(t) = \big[w'(t) - \partial_t B(\gamma,t)\big]/\partial_z B(\gamma,t)$ that recovers the avoidance curve from a prescribed nonvanishing image trajectory $w(t)$.

Ideological Hysteresis
The phenomenon by which a society returns to its original ideological coordinates but on a topologically distinct homotopy sheet, permanently marked by proximity to dangerous attractors encountered along its historical trajectory.

Phase Transition
A moment t^* at which the prescribed image trajectory passes through a critical value of $B(\,\cdot\,, t^*)$, creating a bifurcation in the continuation of the avoidance curve; the mathematical counterpart of a historical turning point.

Safety Index
The magnitude $\Phi(z,t) = |B_{\mathrm{pol}}(z,t)| \in [0,1]$, a real-time indicator of ideological safety: values near 1 indicate equidistance from all dangerous zones in the Poincaré metric; values near 0 signal proximity to at least one attractor.

Winding Number Transport
The continuous evolution of the winding numbers $n_k(t)$ of γ around each moving forbidden point $F_k(t)$, governed by equation (7).

The End